Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/13/2016
PI Name:	Thaxton, Sherry Ph.D.		
Project Title:	Computational Model for Spacecraft/Habitat Volume (Spacecraft Optimization Layout and Volume (SOLV))		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	07/02/2014	**End Date:**	07/01/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Element change to Human Factors & Behavioral Performance; previously Space Human Factors & Habitability (Ed., 1/19/17)		
Key Personnel Changes/Previous PI:	July 2016 report: Removed Joris Guerin (Masters student). Added key personnel: Churlzu Lim, Ph.D. (faculty at University of North Carolina-Charlotte); Sam Wald (Ph.D. student working as contractor intern).		
COI Name (Institution):	Chen, Maijinn M.A. (Lockheed Martin/NASA Johnson Space Center) Hsiang, Simon (University of North Carolina-Charlotte) Myers, Jerry Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A key design challenge for future long-duration exploration missions is determining the appropriate volume of a spacecraft/habitat to accommodate habitability functions and ensure optimal crew health, performance, and safety. Because spacecraft/habitat volume directly drives mass and cost, this information is needed early in the design process. This proposal is in response to the NASA Research Announcement (NRA) NNJ13ZSA002N A.2.i: Computational Modeling and Simulation for Habitat/Vehicle Design and Assessment, and it addresses the Human Research Program (HRP) Program Requirements Document (PRD) Risk of Incompatible Vehicle/Habitat Design. The objective of this proposal is to develop a constraint-driven, optimization-based model that can be used to estimate and evaluate spacecraft/habitat volume. The computational model development will be completed through four Specific Aims: Estimate spacecraft/habitat volume based on mission parameters and constraints, provide layout assumptions for a given volume, assess volumes based on a set of performance metrics, and inform risk characteristics associated with a volume. To accomplish this, the proposed team has been structured to leverage expertise from diverse fields, including architecture and habitation design, human factors engineering, industrial engineering, optimization-based modeling, and simulation. The proposed work will also leverage technical products developed from the HRP-hosted 2012 Habitable Volume Workshop, as well as work performed in the follow-on exploratory project in 2013, including critical task volume estimations and input/output definitions for the computational model. Lessons learned from the development of the Integrated Medical Model (IMM) developed by the Exploration Medical Capability Element (ExMC) of the HRP will also be applied to the proposed work -- lessons ranging from model development approach to compliance with NASA STD 7009, Standard for Models and Simulation. Model validation and verification will be a continuous process occurring throughout model development. The guidelines of NASA-STD-7009 will be followed in establishing parameters and vetting the credibility of the model at all stages of development. The outcome of the proposed work will directly answer to HRP's Risk of Incompatible Vehicle/Habitat Design and the associated Space Human Factors Engineering (SHFE) SHFE-HAB-09 Gap on technologies, tools, and methods for data collection, modeling, and analysis for design and assessment of vehicles/habitats. A computational model for spacecraft/habitat volume will be an invaluable tool for designers, mission planners, integrators, and evaluators who are shaping space habitats and working toward a truly habitable environment for future long-duration exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth industries that are concerned with habitability in confined environments for long durations (e.g., shipping, submarines, oil and gas rigs, Antarctic research stations) may benefit from the task-based approach in development for determining overall volume needs.
Task Progress:	Please note that funding to Dr. Simon Hsiang, a critical co-investigator for the development of the Spacecraft Optimization Layout and Volume (SOLV) model, was unavailable to him for 9 months following his transfer to a new university. As such, progress for this year was limited and a no-cost extension will result in an end date in FY18. However, interim work took place as summarized here. At the Year 1 End-of-Year Review (July 2015), SOLV's Key Driving Requirements (KDRs) were peer-reviewed and concurred. Line-by-line reviews resulted in updates to the requirement text and the proposed verification methods and success criteria. As part of the outcome of this review, the SOLV team worked closely with NASA to focus on the documentation of task volume assumptions, which will be used as inputs to SOLV; and to develop a process to represent varying potential task volumes using a strategy referred to as the gradient cuboid method. In efforts to define the variables and metrics that will drive SOLV, the team began investigating the use of the Analytical Hierarchy Process (AHP) to systematically assign values and priorities to habitat volume drivers. AHP will be used as a decision support technique to identify factors with the greatest probable influence and, used in conjunction with Choquet integrals, will feed the model's output of a scorecard that provides an at-a-glance assessment of overall costs and benefits associated with layouts generated by SOLV. In addition to efforts to refine the algorithm approach, the SOLV team concentrated on ensuring proper documentation of the modeling efforts. The team finalized a matrix mapping planned SOLV documentation to requirements for NASA-STD-7009. To begin documentation efforts, the team drafted an initial version of a Verification and Validation (V&V) plan, which will be updated throughout model development to reflect the current state of V&V. The team also drafted an outline for the SOLV User Guide, which will document details such as assumptions regarding user qualifications and problem reporting. When funding became available to the University of North Carolina-Charlotte (UNC-C) team members, work resumed on code based bin-packing algorithms that will be used to generate layouts. Gradient cuboid representations of task volumes will feed directly into the bin-packing code, which, based on defined constraints, will generate multiple potential task layouts that will in turn be assessed using criteria driven out by the AHP and Choquet integral process. The bin-packing algorithm currently assesses space in two dimensions, with plans in the near future to expand to three dimensions.
Bibliography Type:	Description: (Last Updated: 06/01/2017)
Abstracts for Journals and Proceedings	Thaxton SS, Chen M, Hsiang S, Myers J. "Spacecraft Optimization Layout and Volume (SOLV): Development of a Model to Assess Spacecraft/Habitat Volume." Presented at 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016